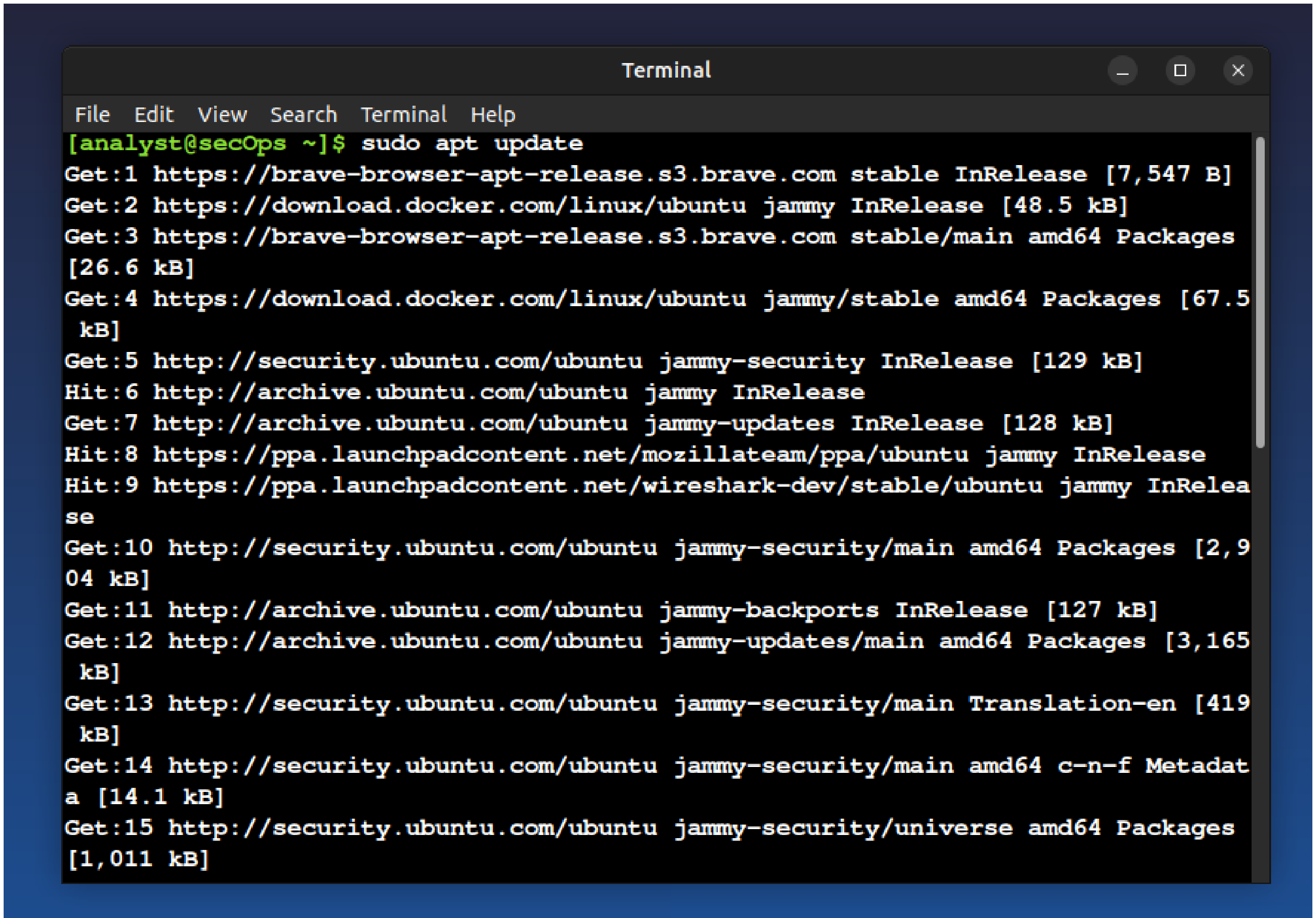
DNS Analysis & Troubleshooting Using Wireshark

Graded Lab Project

Ubuntu & Wireshark

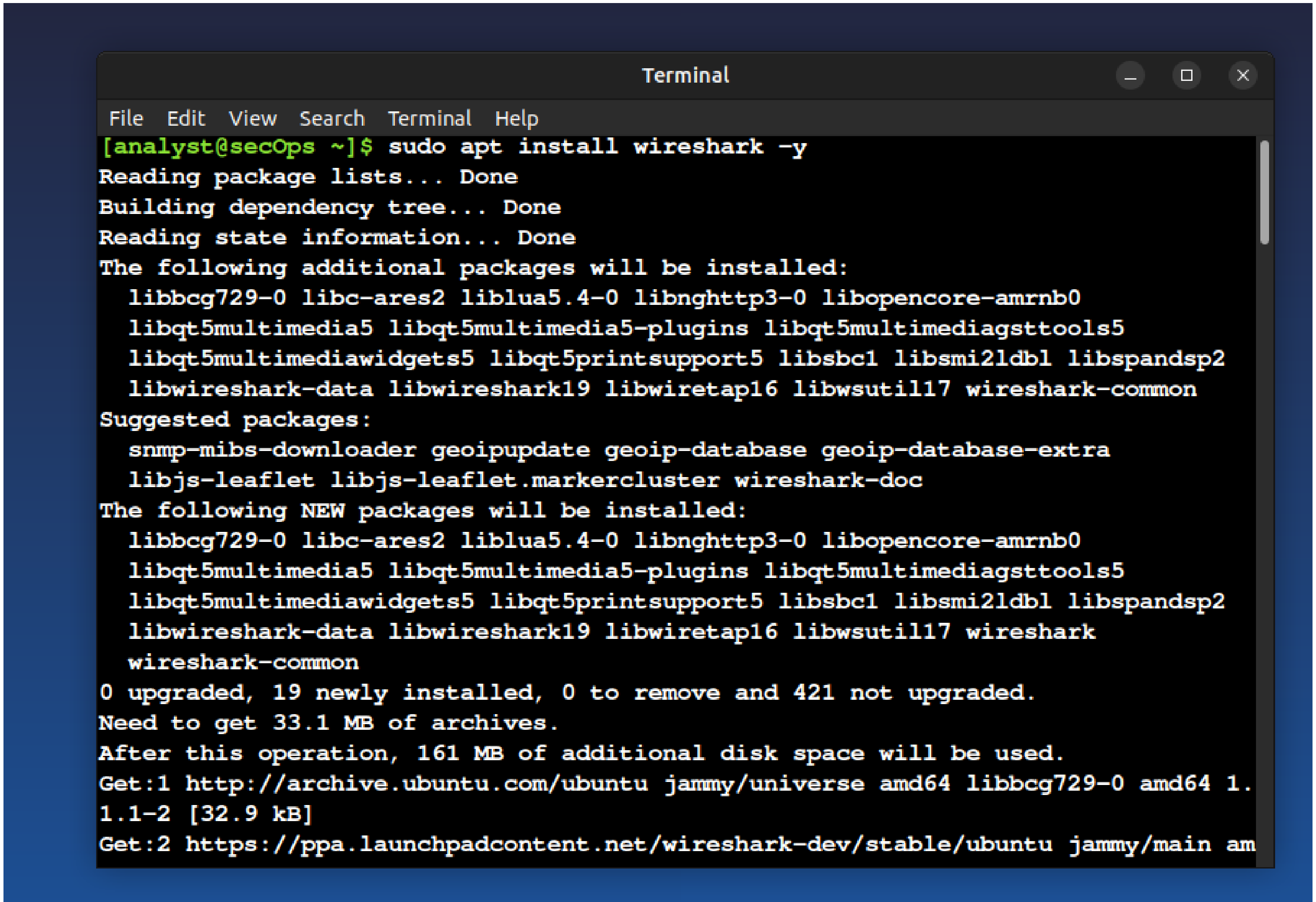
Step 1

Command: **sudo apt update**
Explanation: Updates the package list and checks for available software updates.



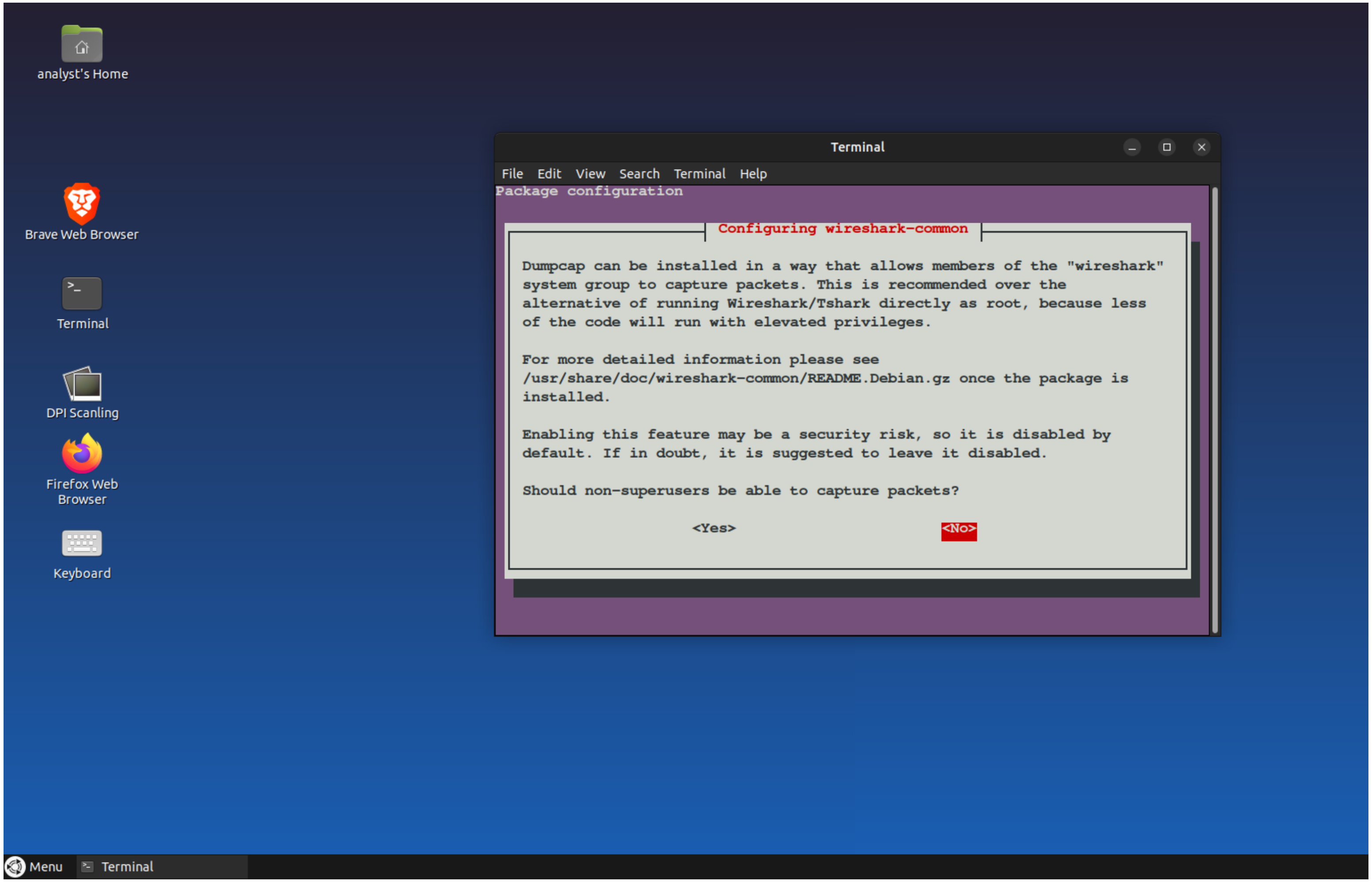
Step 2

Command: **sudo apt install wireshark -y**
Explanation: Installs Wireshark on the Ubuntu system.



Step 3

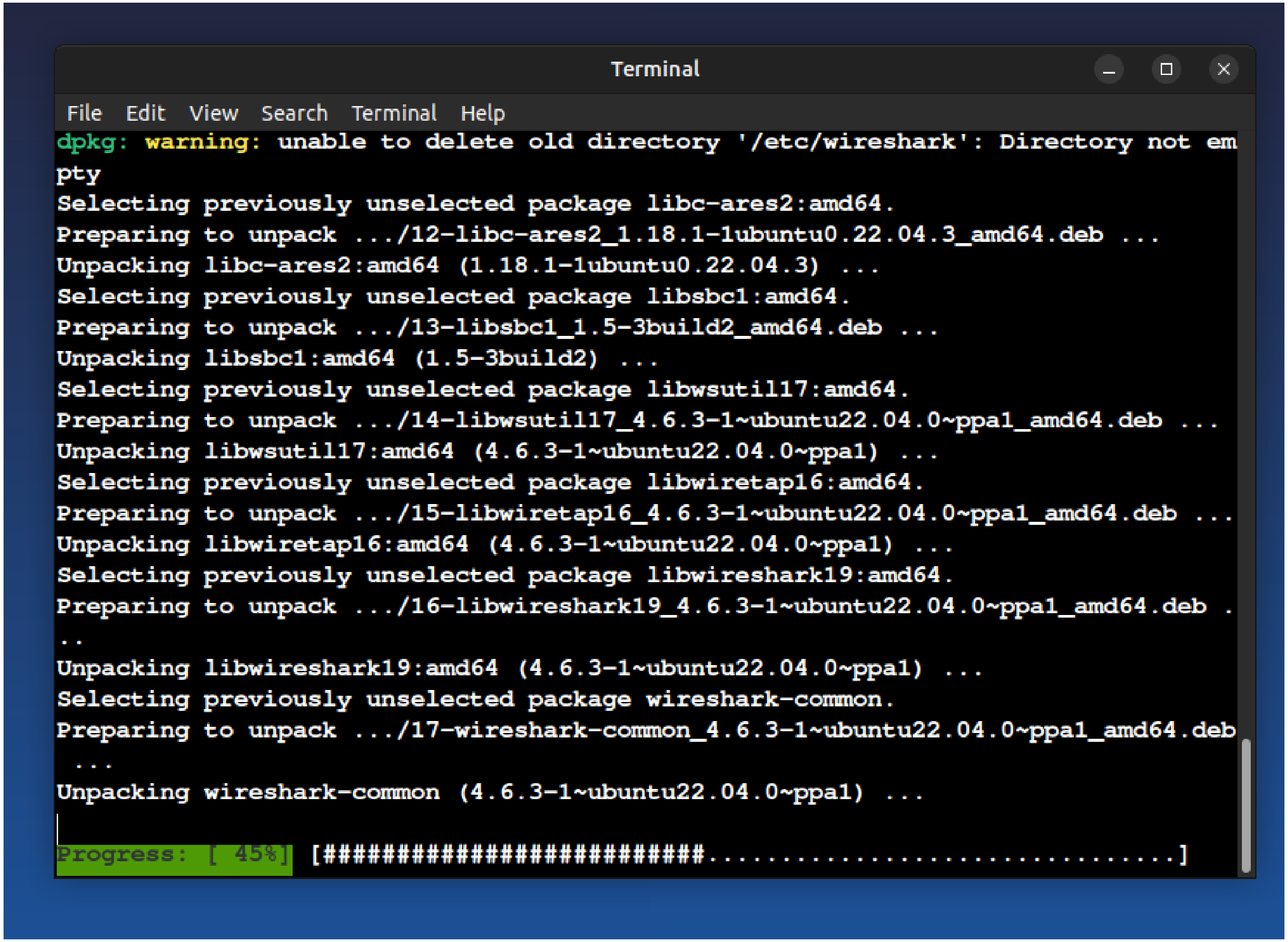
Command: ***Wireshark configuration prompt***
Explanation: Allows non-root users to capture packets by selecting Yes.



Step 4

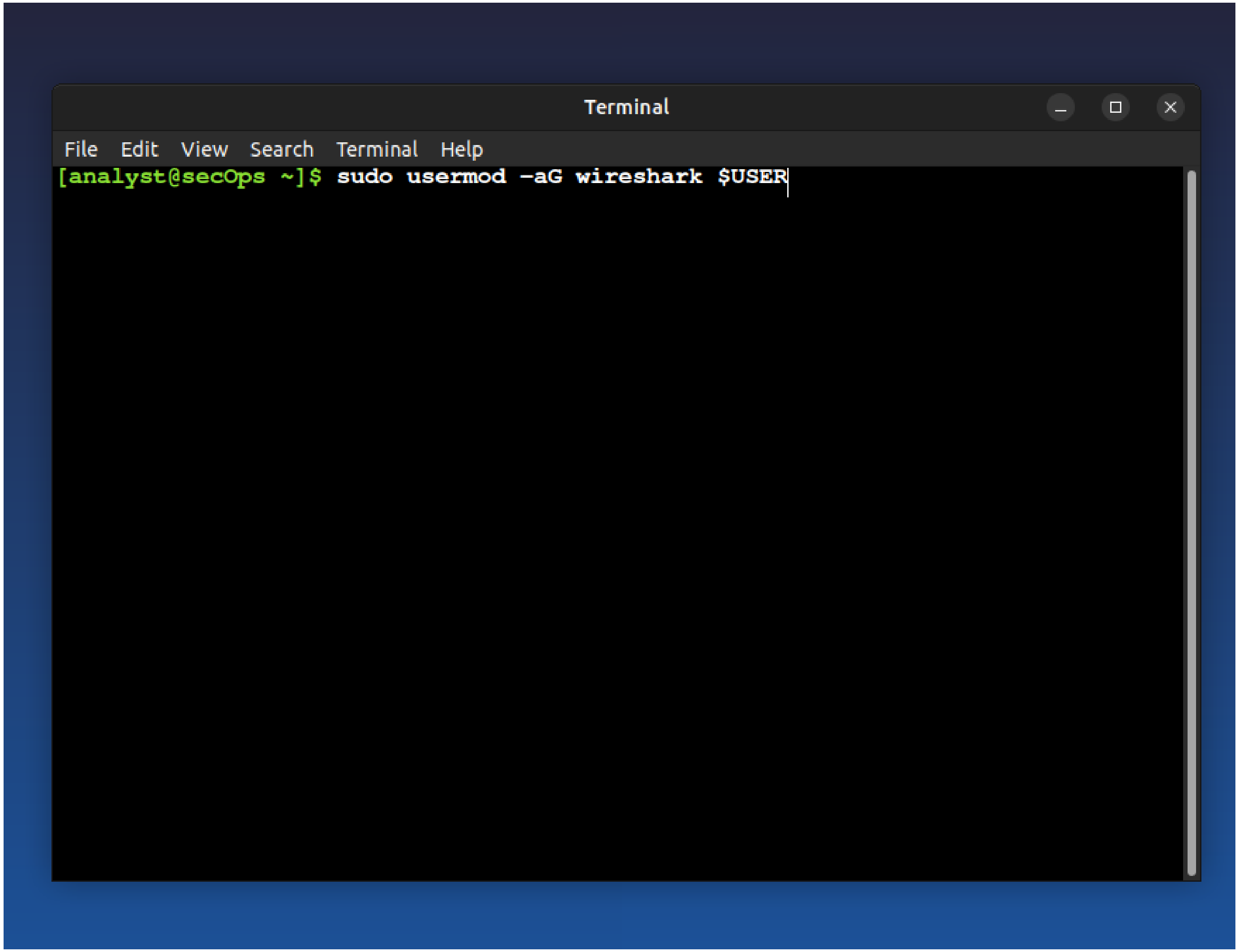
Command: Installing dependencies

Explanation: Wireshark and required libraries are being installed.



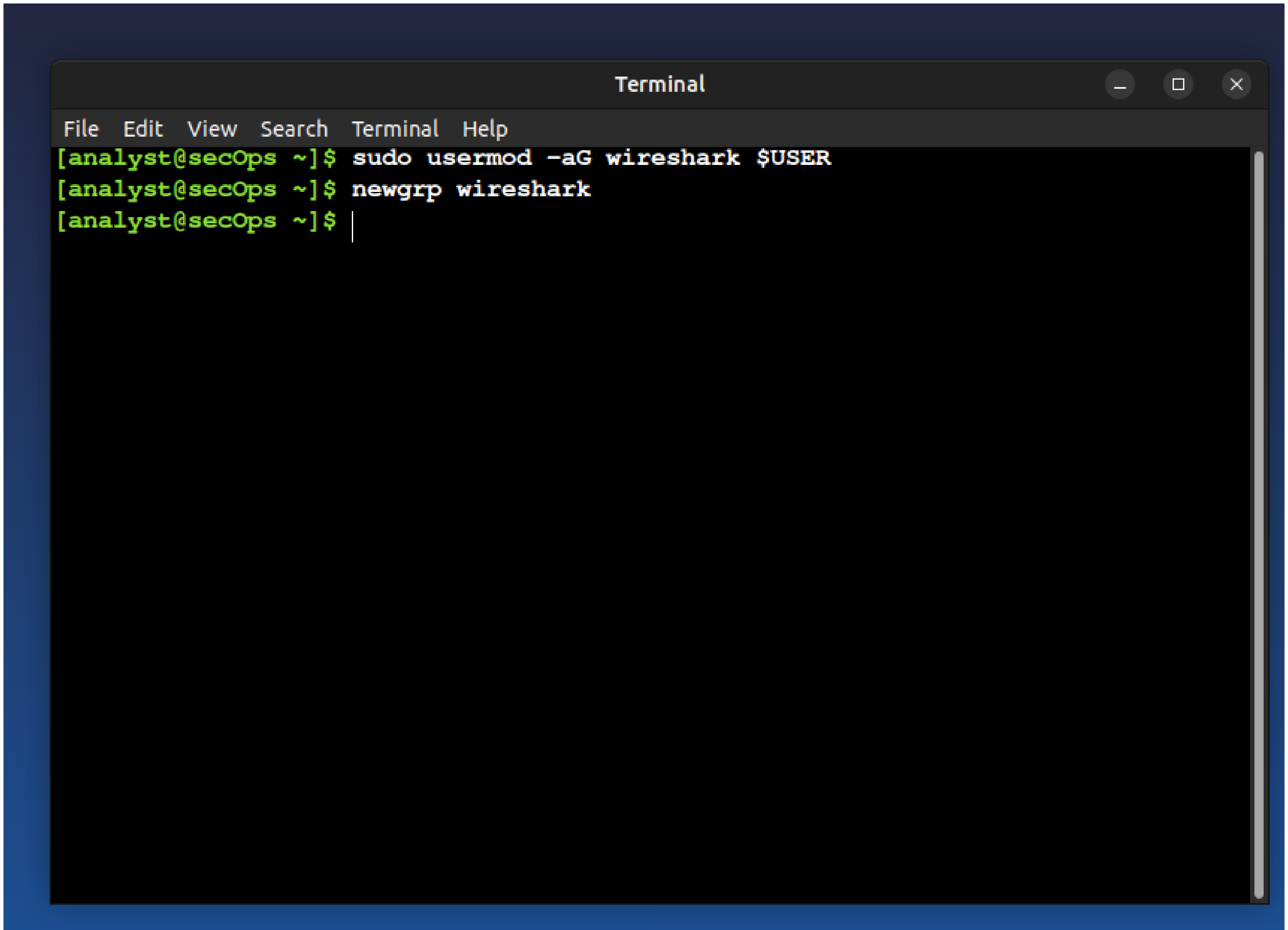
Step 5

Command: **sudo usermod -aG wireshark $USER**
Explanation: Adds the current user to the wireshark group.



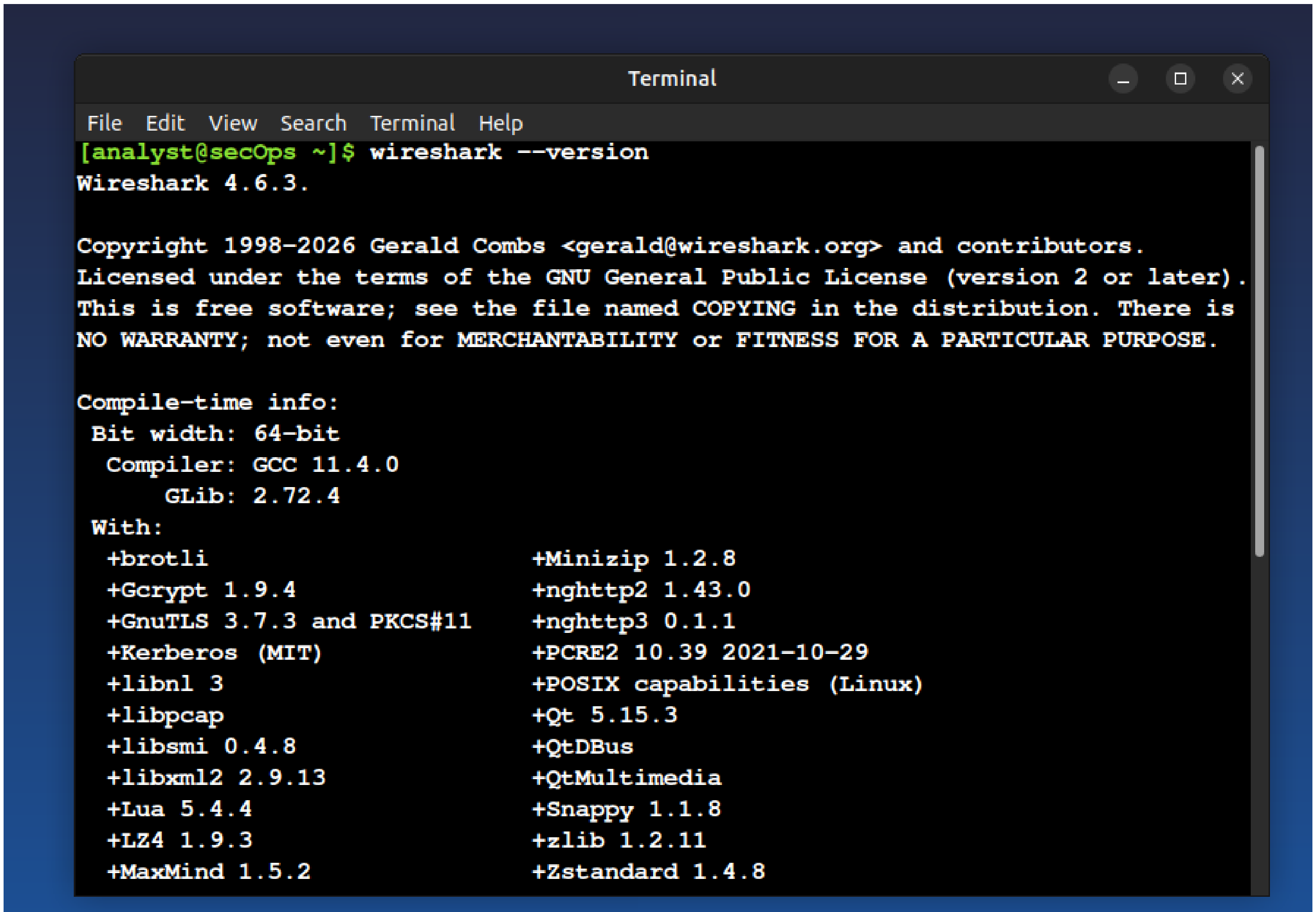
Step 6

Command: ***newgrp wireshark***
Explanation: Applies the group change without logging out.



Step 7

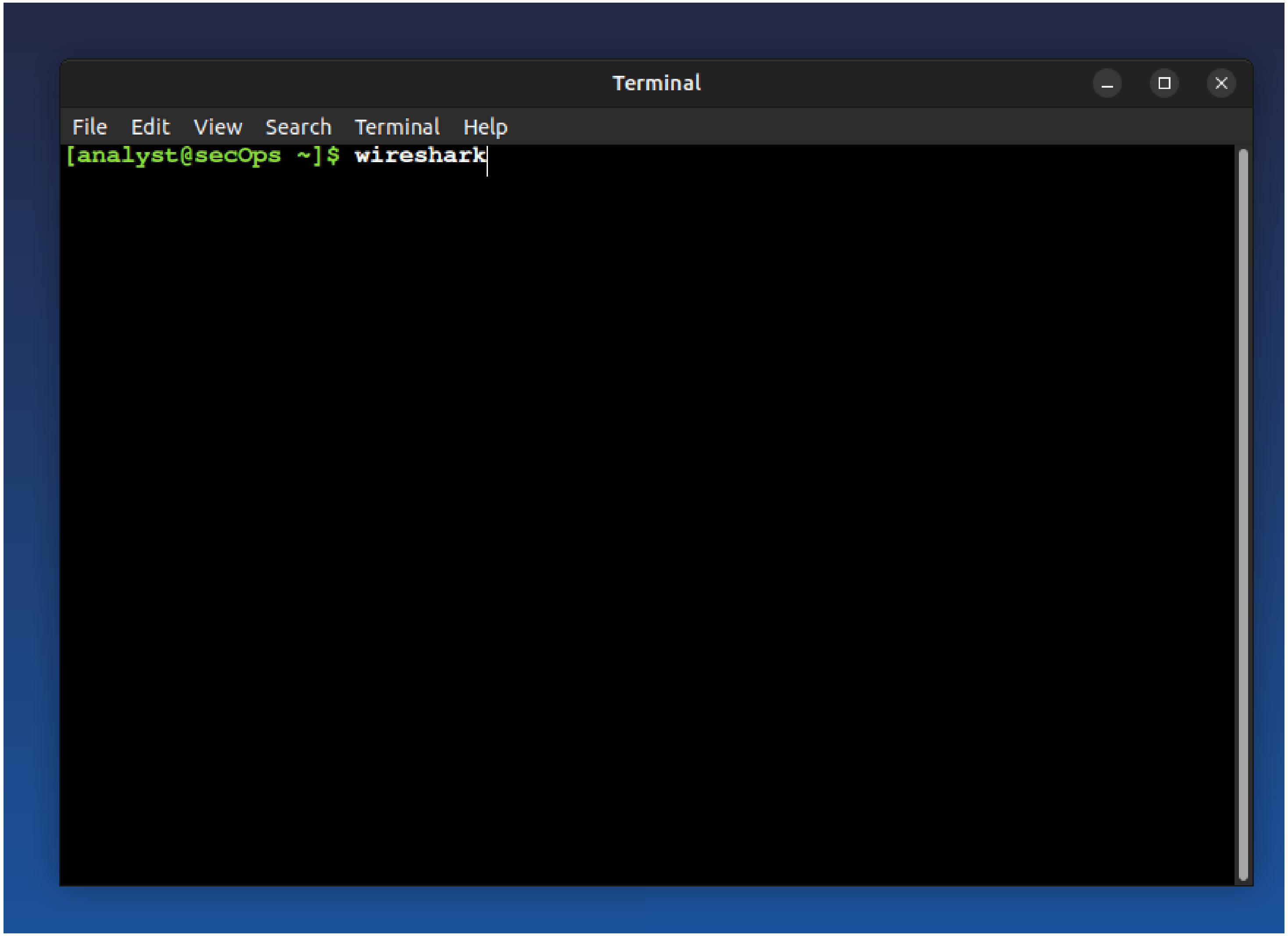
Command: **wireshark --version**
Explanation: Verifies that Wireshark is installed correctly.



Step 8

Command: ***wireshark***

Explanation: Launches the Wireshark application.



Step 9

Command: ***cat /etc/resolv.conf***

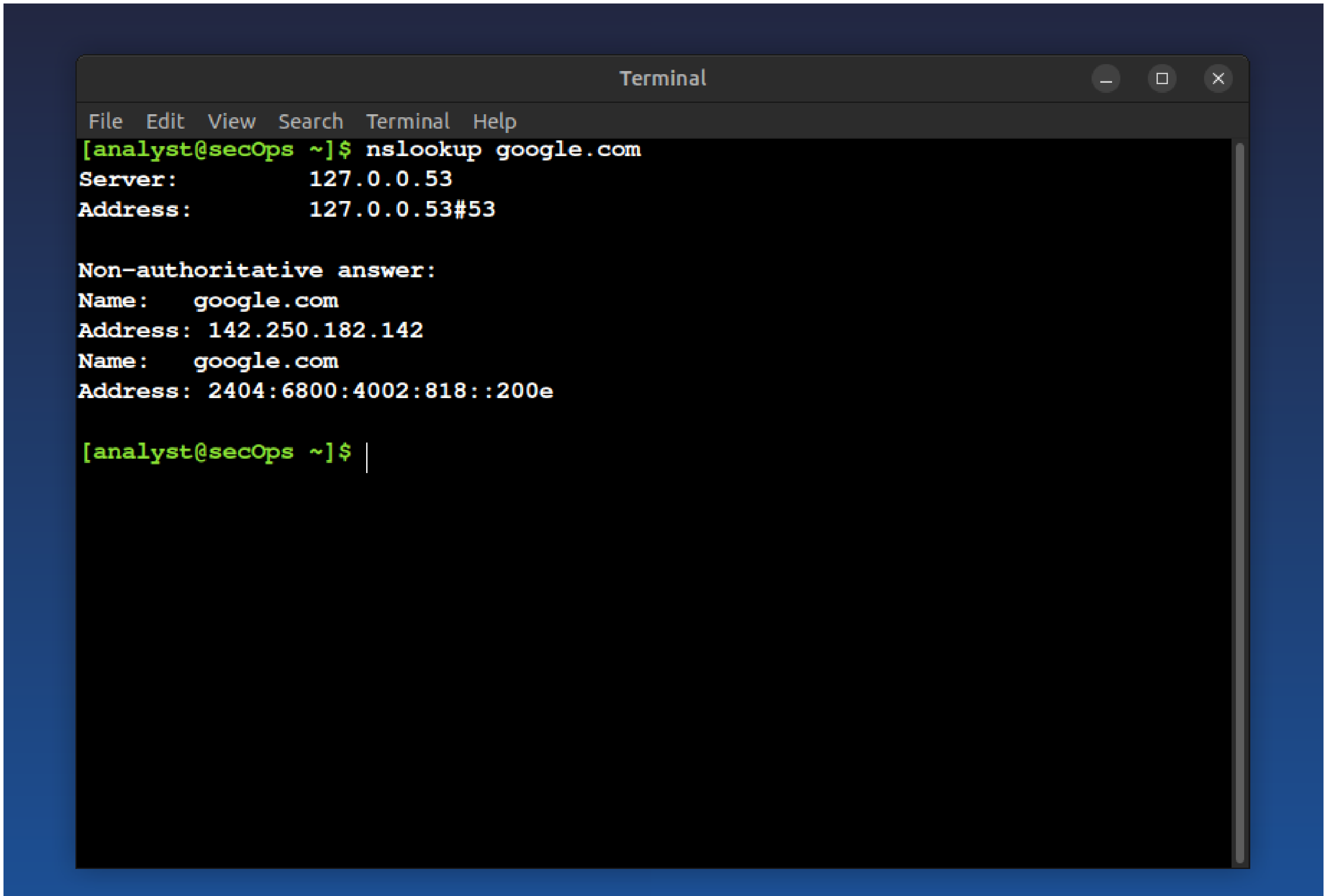
Explanation: Shows the current DNS configuration used by the system.

```
[analyst@secOps ~]$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(
8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 127.0.0.53
options edns0 trust-ad
```

Step 10

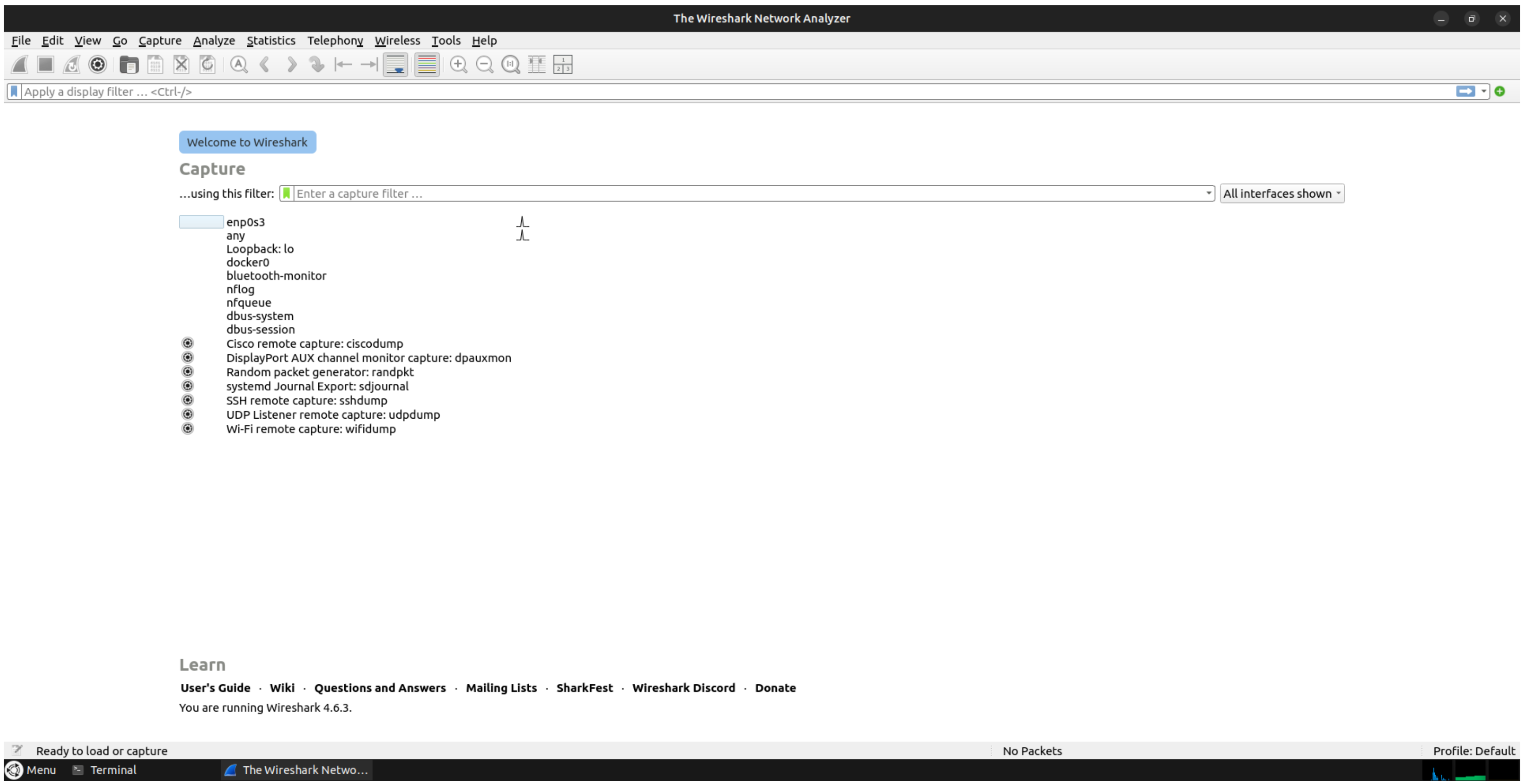
Command: **nslookup google.com**
Explanation: Tests DNS resolution for a valid domain.



Step 11

Command: **Open Wireshark**

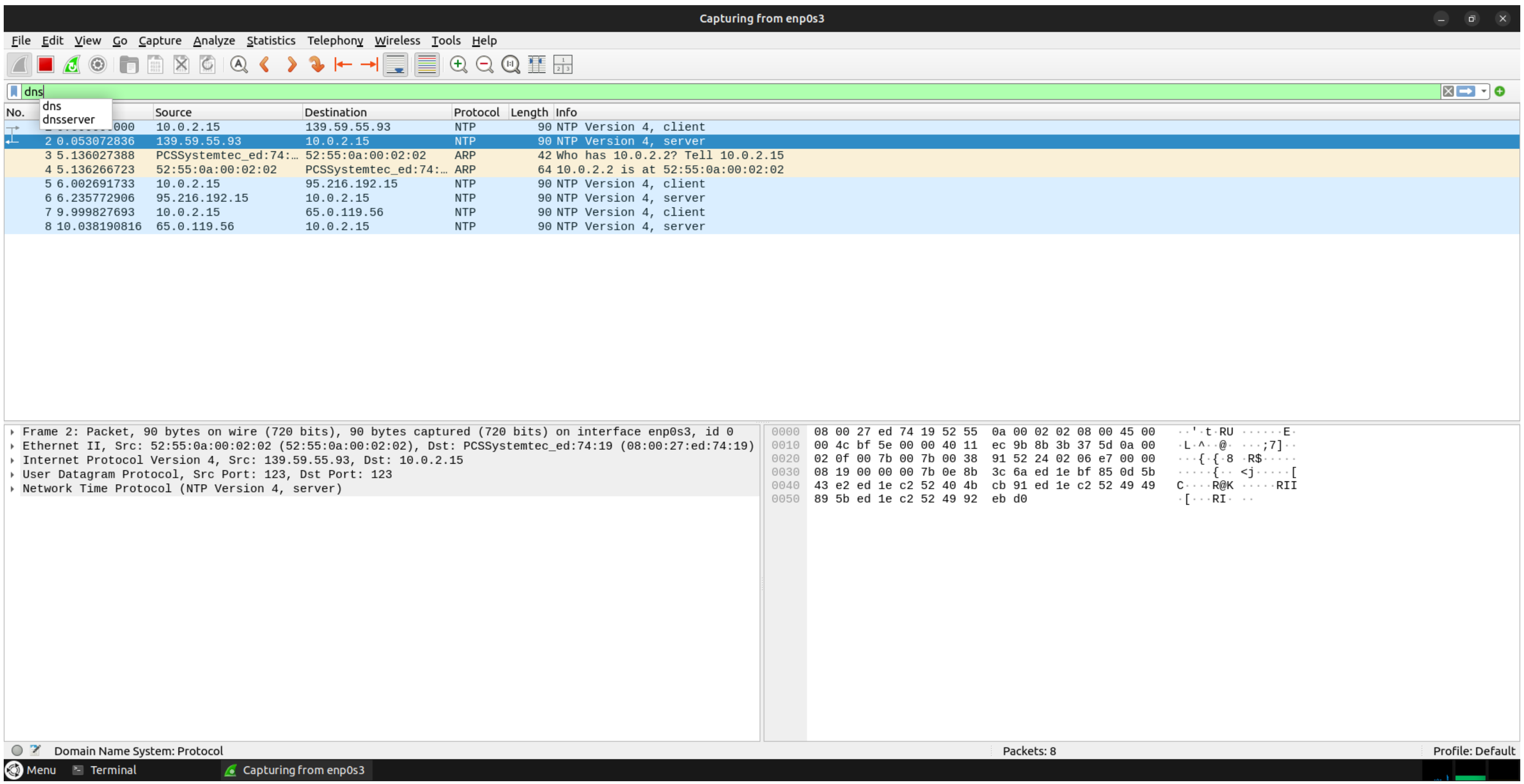
Explanation: Wireshark interface is opened for packet capture.



Step 12

Command: **Apply dns filter**

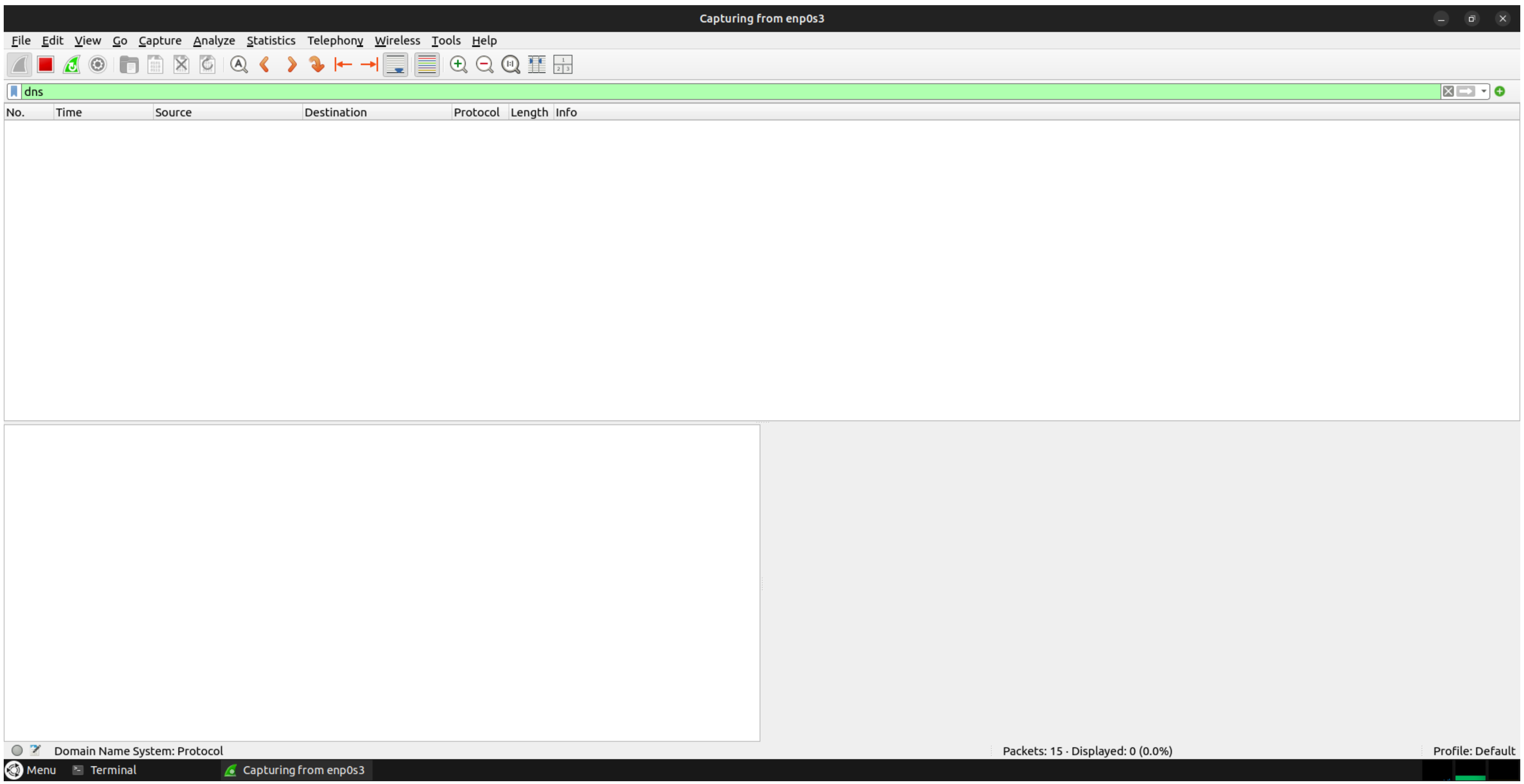
Explanation: Filters only DNS packets in Wireshark.



Step 13

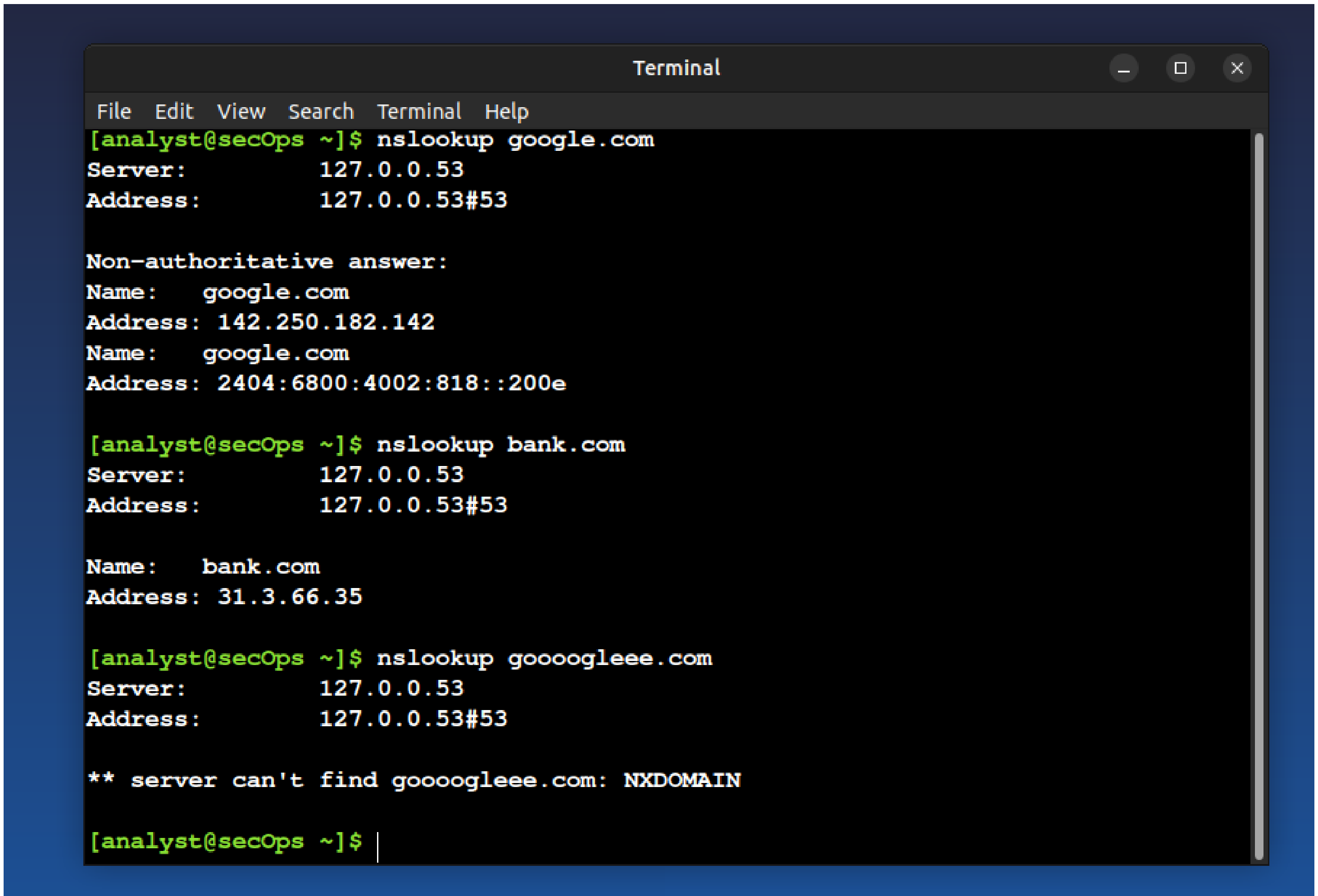
Command: **Start capture**

Explanation: Wireshark starts capturing DNS traffic.



Step 14

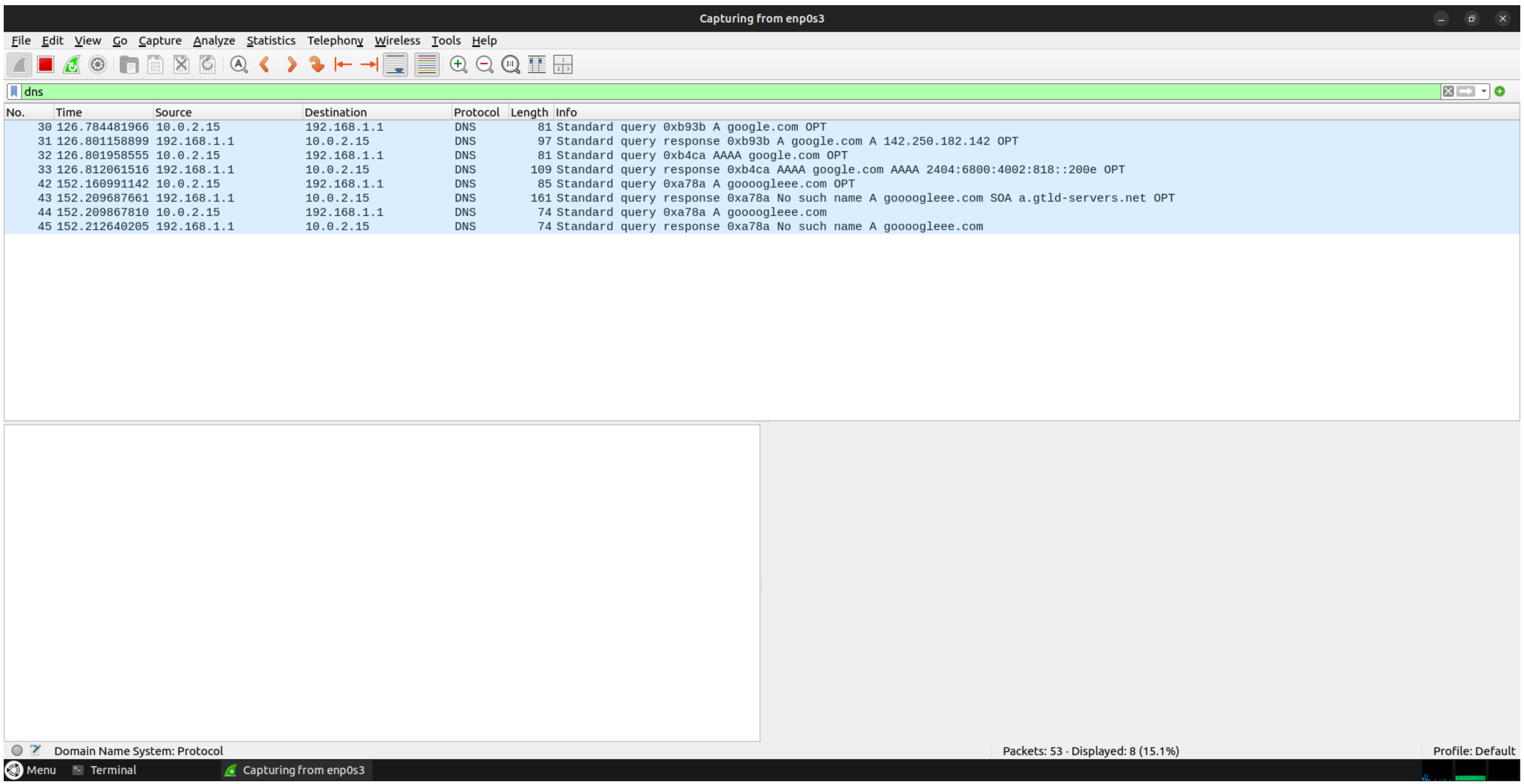
Command: ***Generate DNS traffic***
Explanation: DNS queries are generated using nslookup.



Step 15

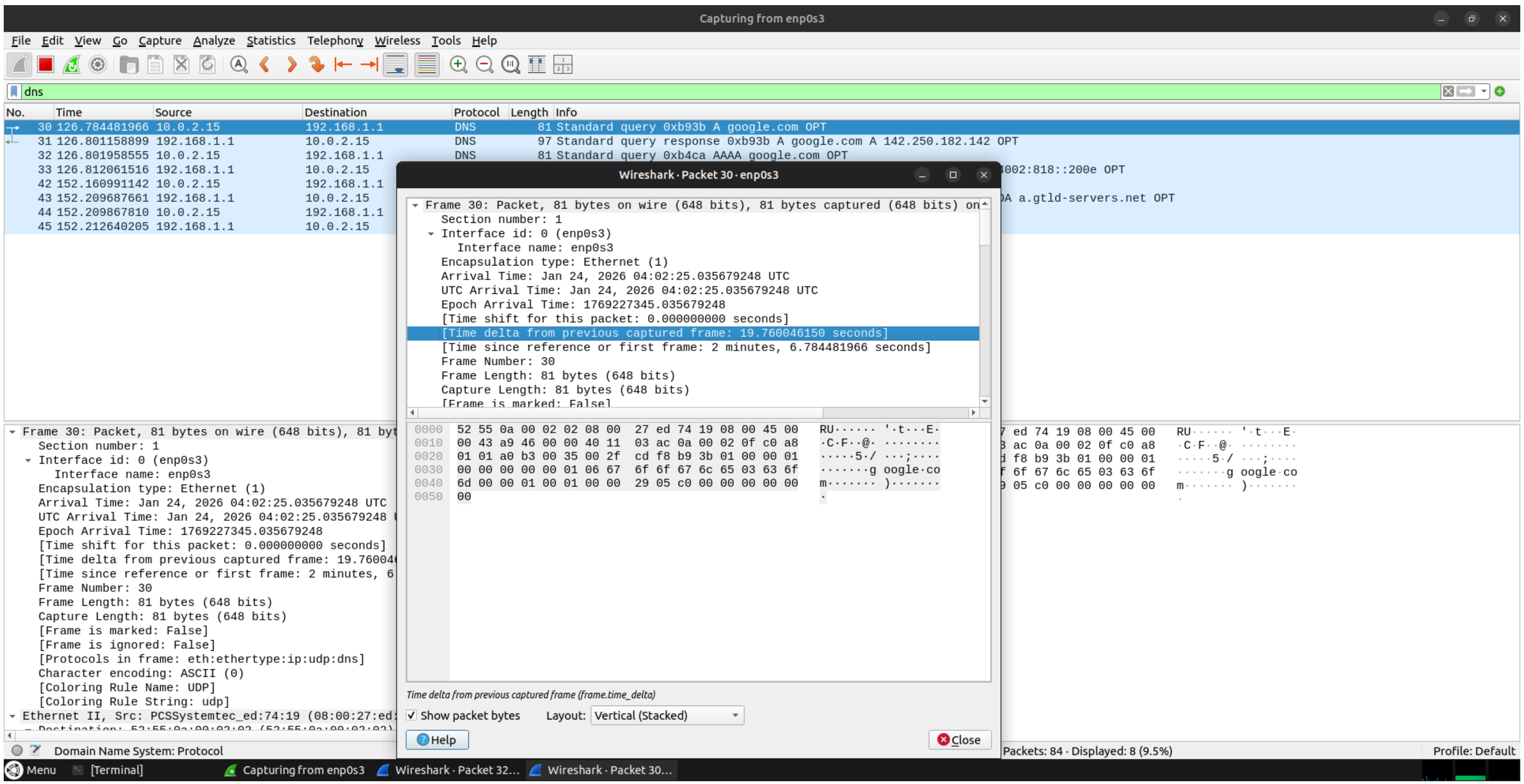
Command: **Analyze DNS packets**

Explanation: DNS requests and responses are visible in Wireshark.



Step 16

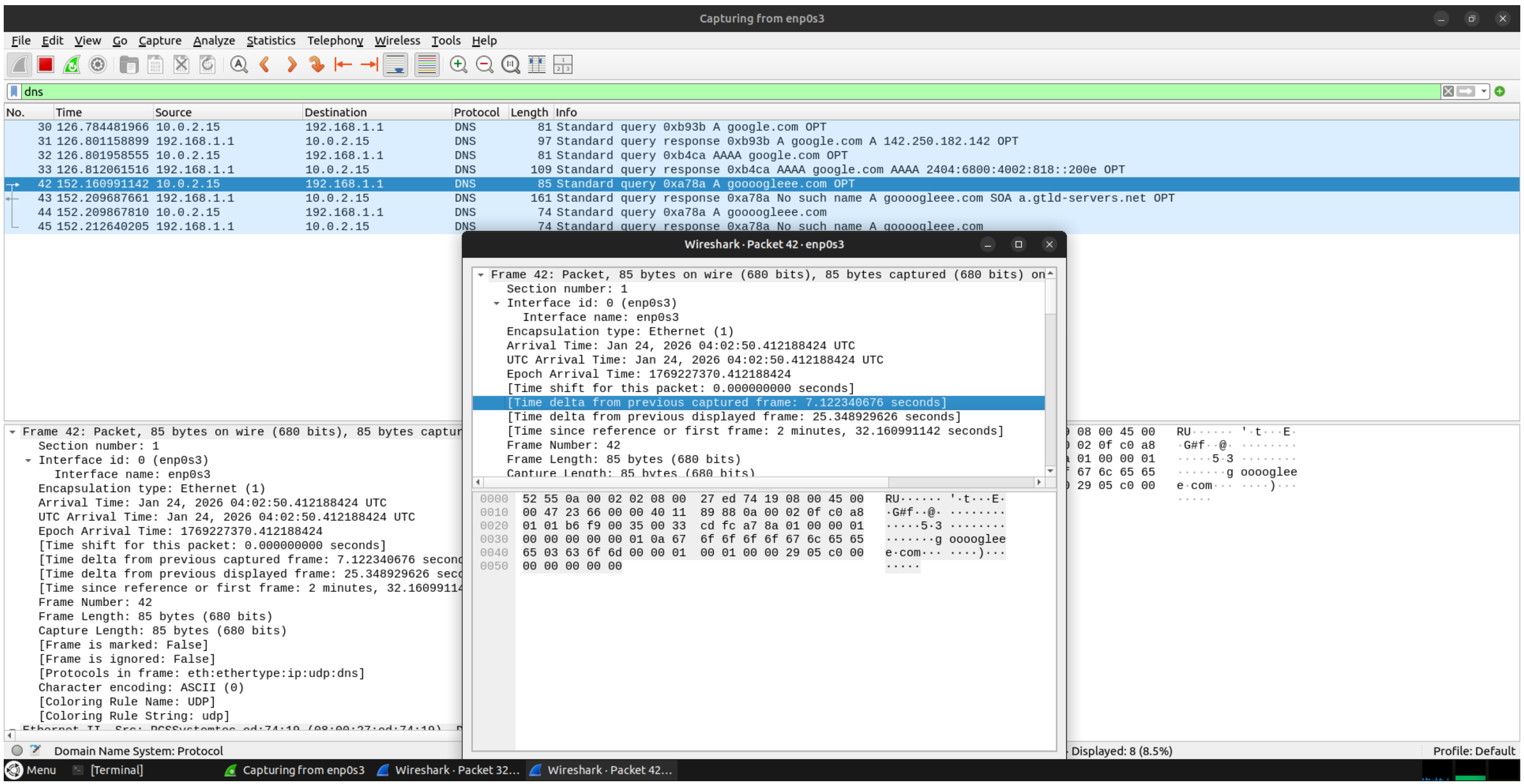
Command: Check response time (***google.com***)
Explanation: DNS response is fast due to reachable DNS server.



Step 17

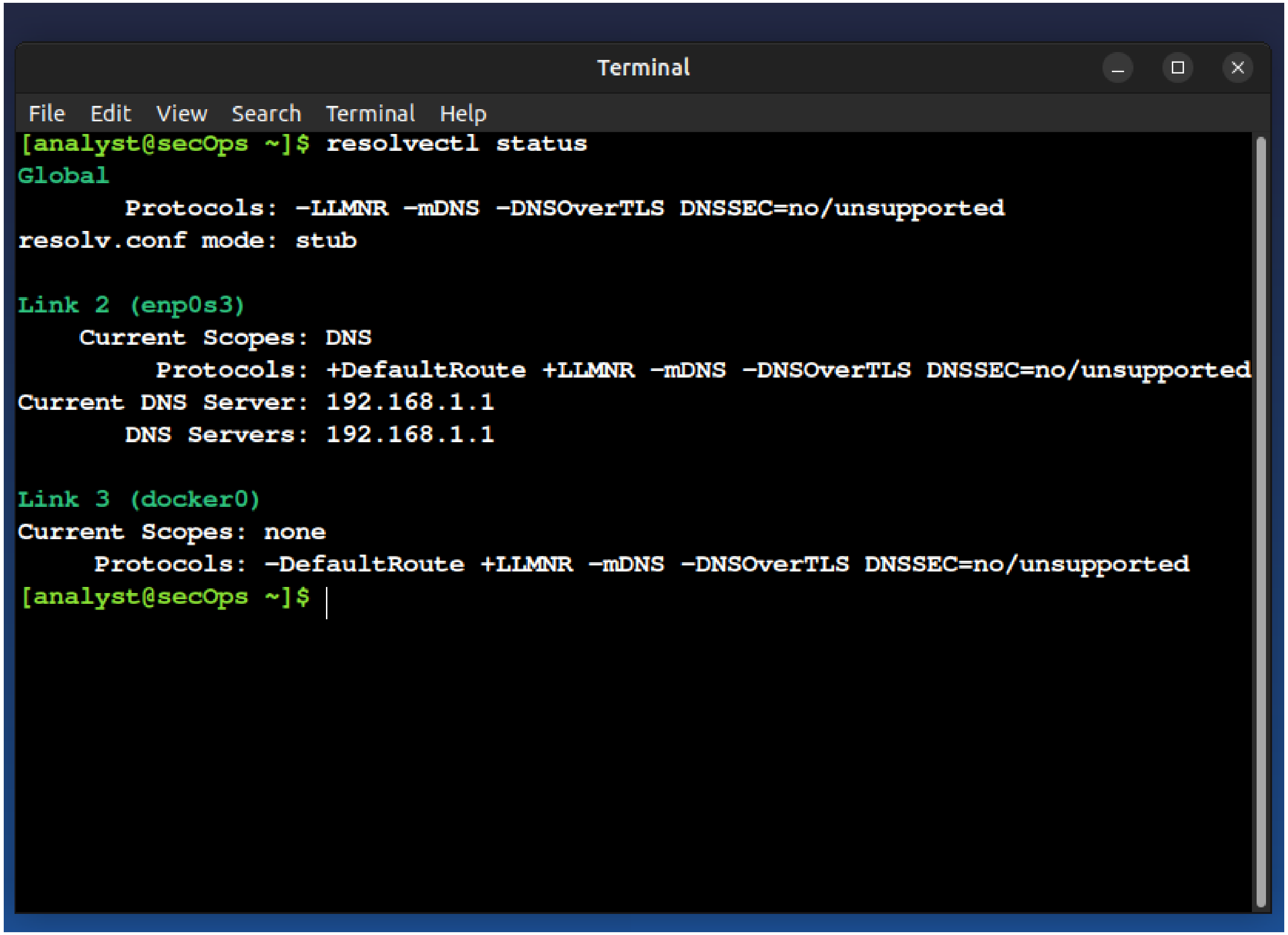
Command: Check **NXDOMAIN**

Explanation: Invalid domain returns NXDOMAIN response.



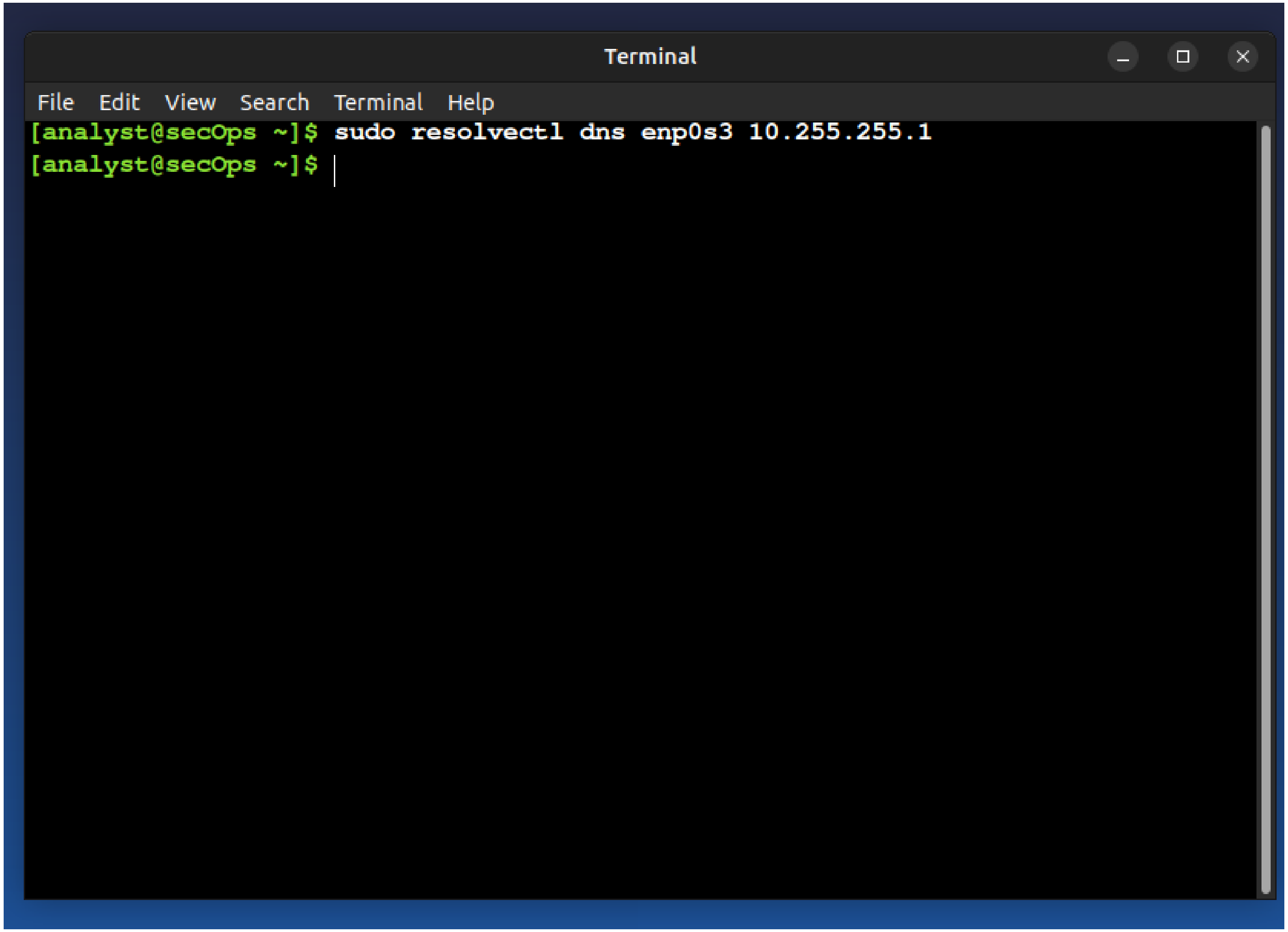
Step 18

Command: **resolvectl status**
Explanation: Shows current DNS server configuration.



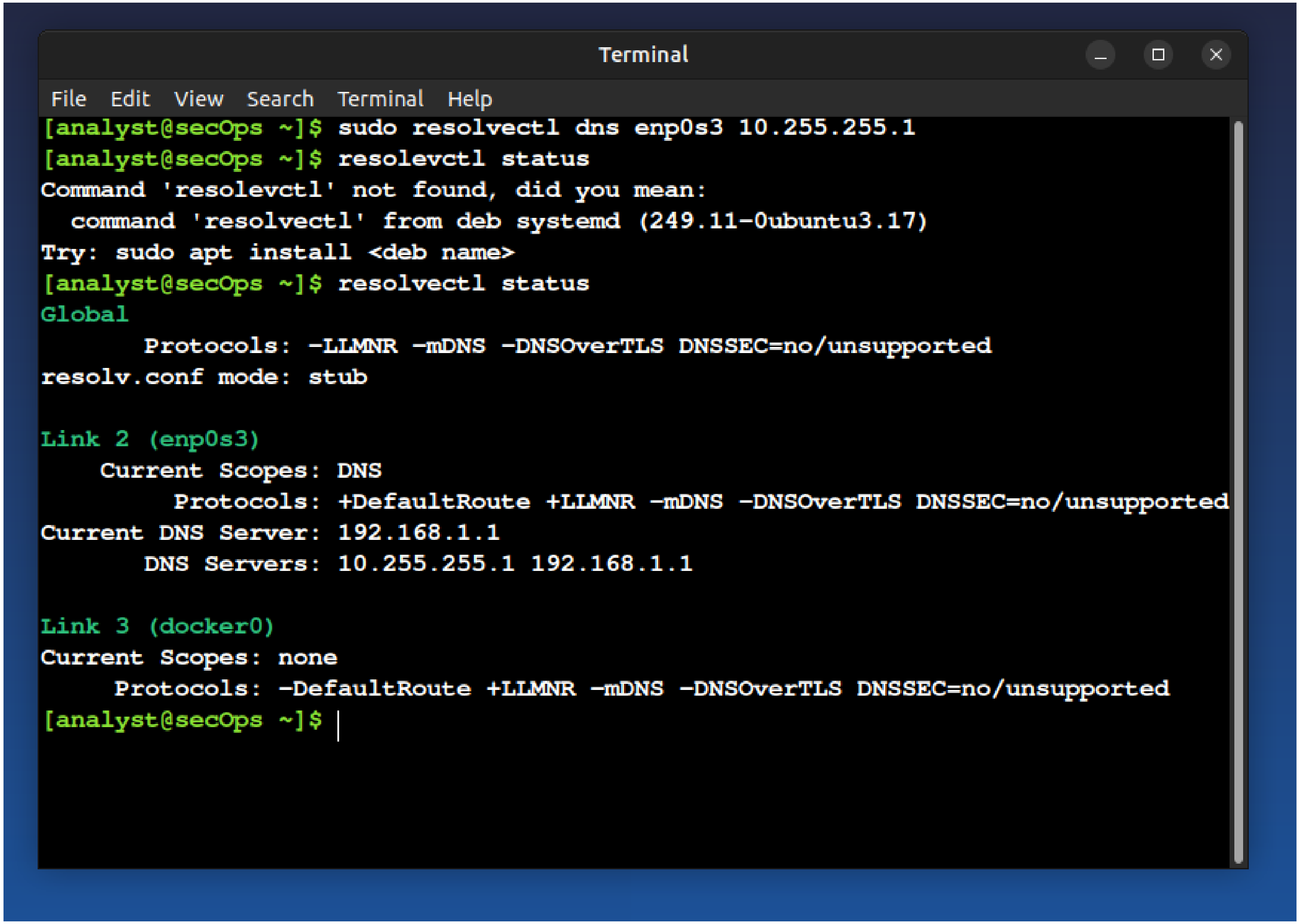
Step 19

Command: **Set bad DNS (10.255.255.1)**
Explanation: System DNS is changed to an unreachable server.



Step 20

Test DNS again. Command: **resolvectl status**
Explanation: DNS resolution becomes slow due to timeout.

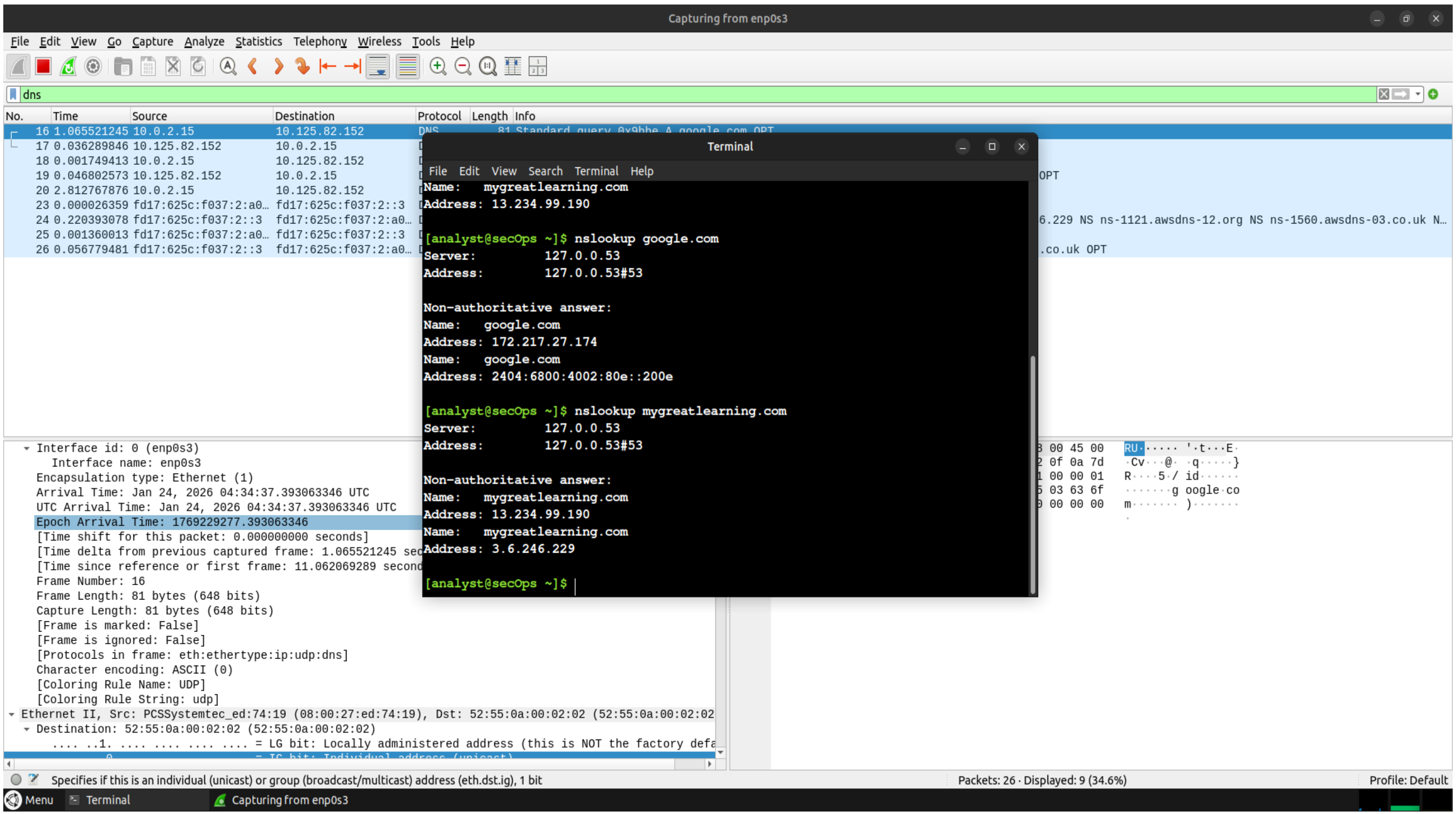


Step 21

Observe delay

Explanation: Wireshark shows delayed DNS response.

In my ubuntu lab I noticed that the nslookup google.com took 2 seconds in terminal to show details and nslookup mygreatlearning.com took 4 to 5 seconds to show details in terminal.

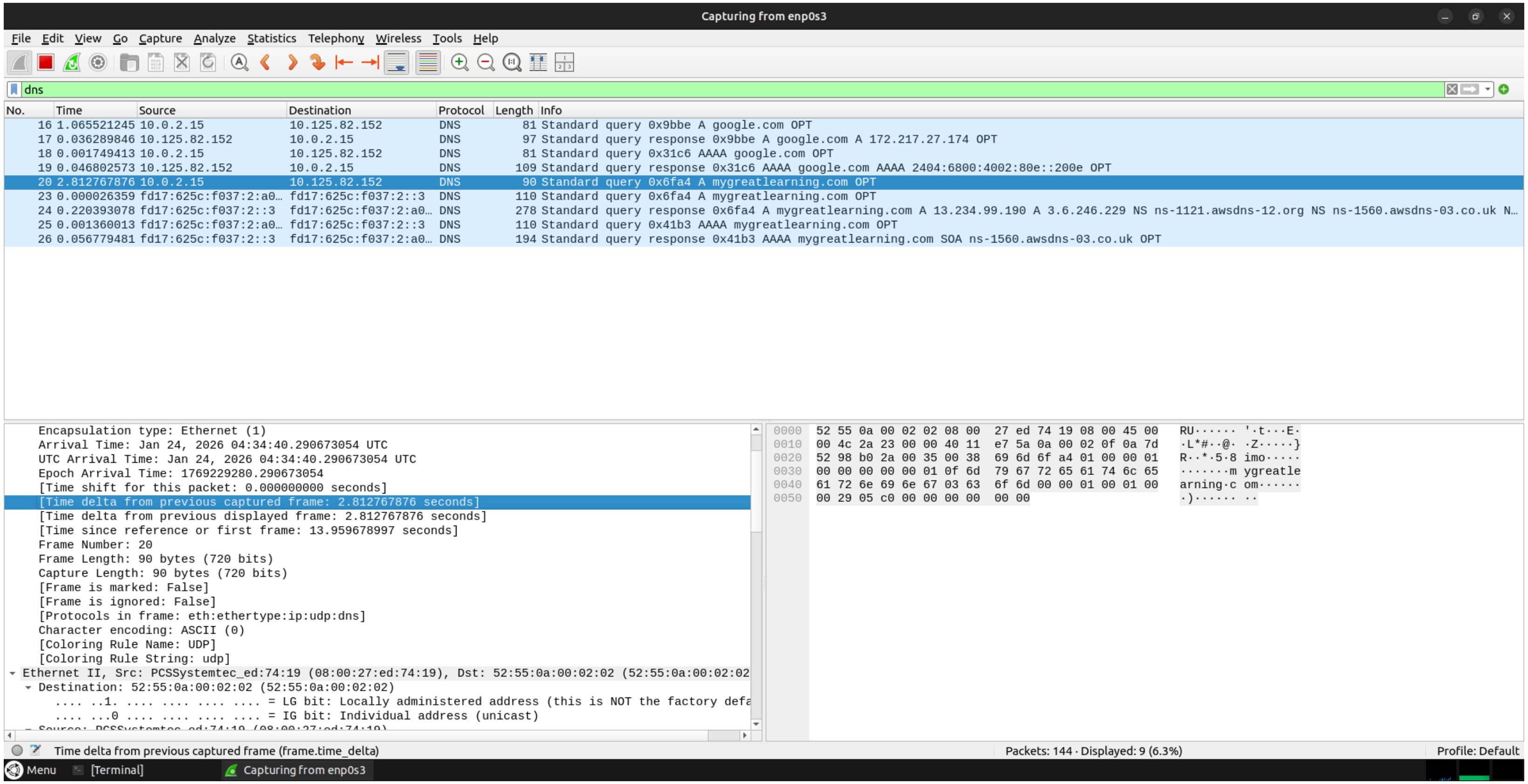


Step 22

DNS request failed

Explanation: Initial DNS server does not respond.

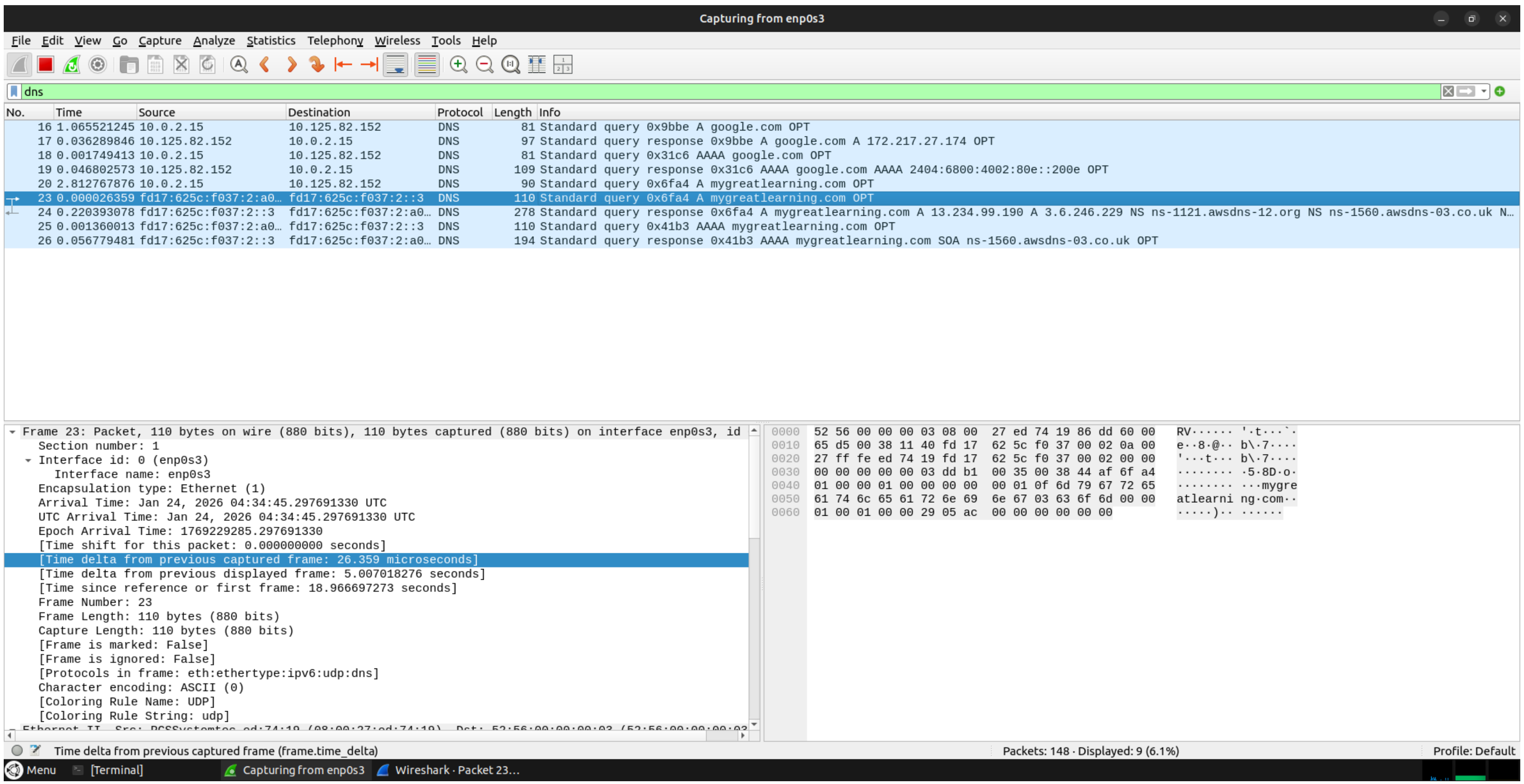
First it tried with 10.255.255.1 and it waited for aprox. 5 seconds and then again tried with original DNS server i:e 192.168.1.1



Step 23

Retry with original DNS

Explanation: System retries using secondary DNS server i:e 192.168.1.1

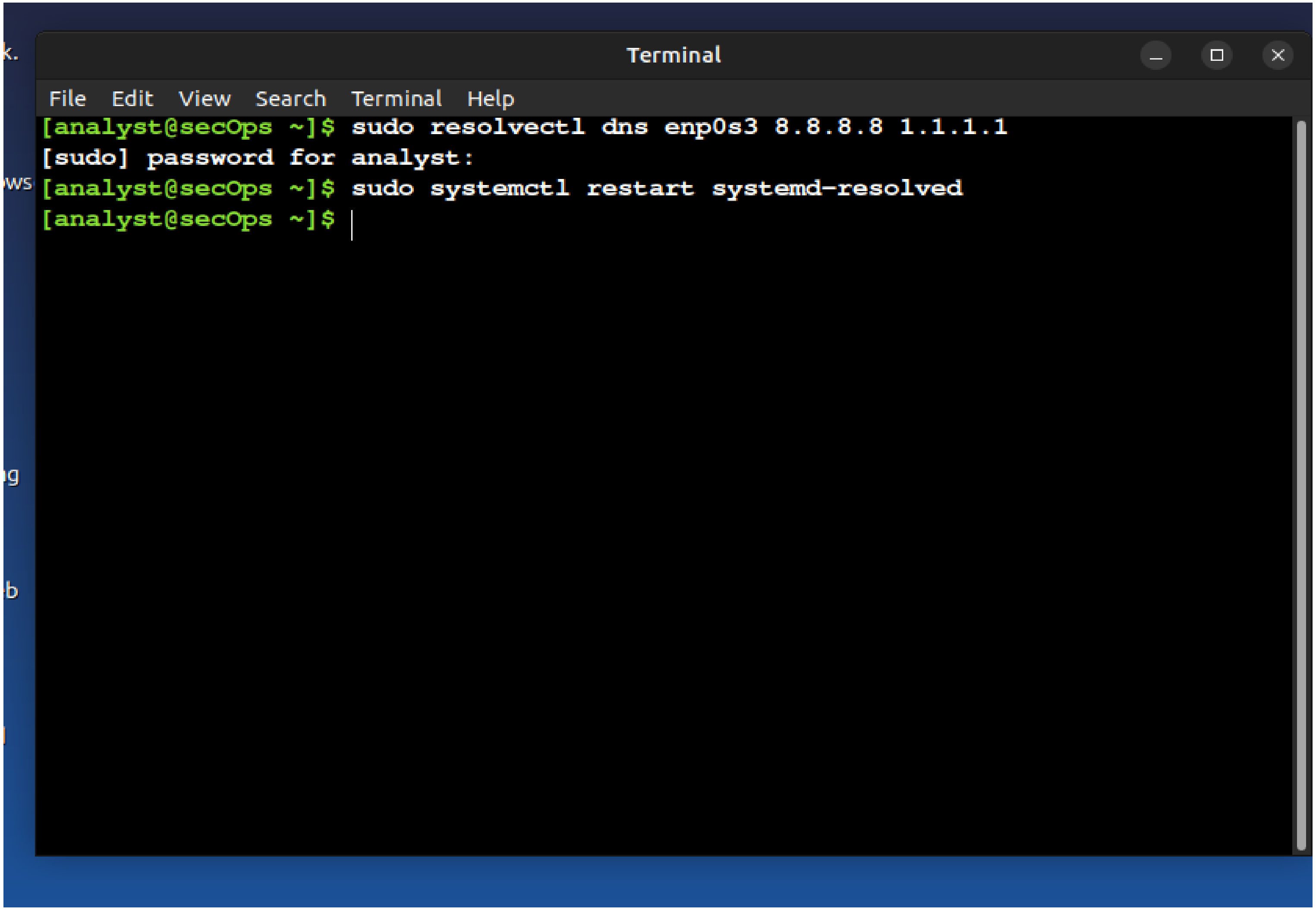


Step 24

Command: **sudo resolvectl dns enp0s3 8.8.8.8 1.1.1.1**

Explanation: DNS is changed to public servers (8.8.8.8, 1.1.1.1).

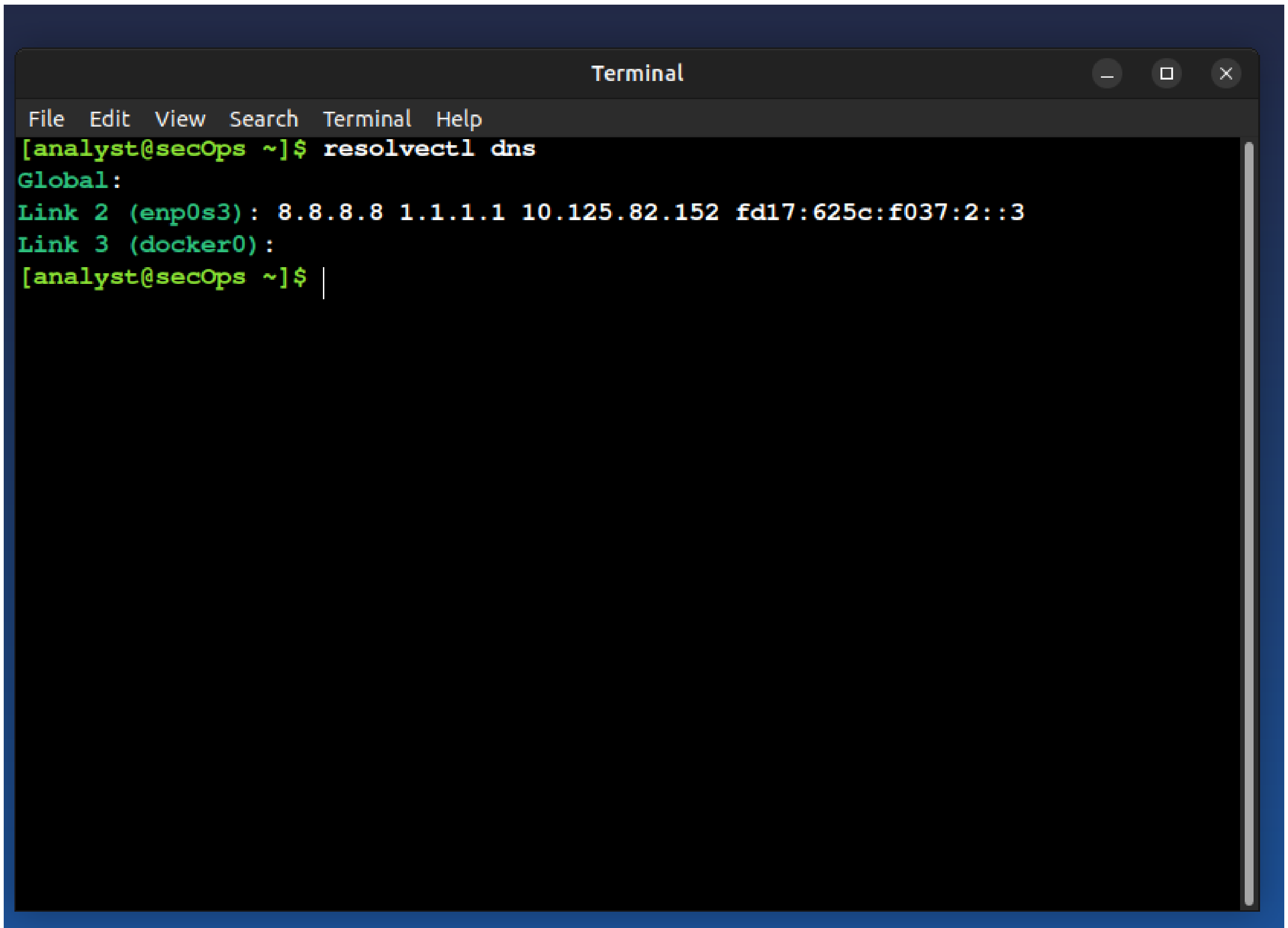
Google = 8.8.8.8 and cloudfare = 1.1.1.1



Step 25

Command: **Restart systemd-resolved**

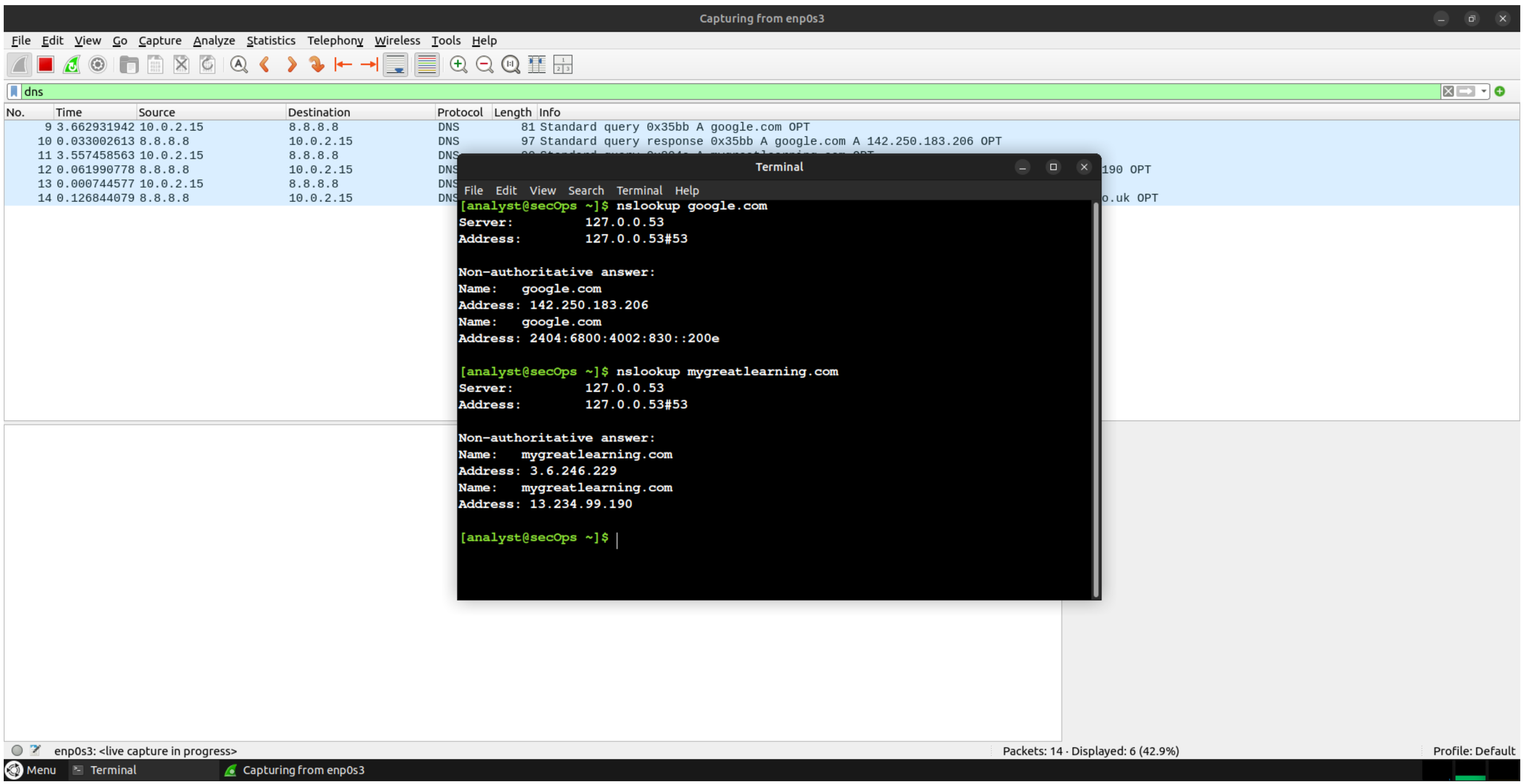
Explanation: Applies the DNS configuration changes.



Step 26

Test DNS after fix

Explanation: DNS resolution becomes fast again.



Step 27

Verify timing

Explanation: Wireshark confirms normal DNS response time.